

Event Boundaries and Memory in the Mnemonic Similarity Task

BRSM 2026 - Report 2: Advanced Inferential Analysis

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1 Introduction and Background

Human experiences are not in a continuous stream, instead the brain automatically segments the ongoing perceptual changes into discrete events to optimize this structural encoding and the situational awareness. Event boundaries occur whenever there is a perceptual or situational change like moving between the rooms or say experiencing a temporal shift or a scene change in a visual manner (Zacks, Speer, Swallow, Braver, Reynolds, 2007). Event segmentation theory suggests that these event boundaries trigger a cognitive update in working memory, which briefly impairs the encoding accuracy for the stimuli immediately after post-boundary (Swallow, Zacks, Abrams, 2009).

1.1 Summary of Initial Insights (Report 1)

Human experiences are naturally divided into discrete events and the prior research in this field shows that event boundaries do play an important role in shaping the memory. Studies have also found the recognition for the items after the boundary are weaker than the middle items which suggests that there is disruption at points which have some perceptual or contextual change.

We have analysed the data from 160 participants across three different cognitive load conditions (Item Only, Task Only, Both) using the Mnemonic Similarity Task (MST). MST measures recognition (REC) and mnemonic discrimination (LDI) using targets, lures, and foils.

This study examines how the memory performance is affected by event boundaries based on cognitive load and perceptual similarity. It also shows how these event boundaries can disrupt the recognition and also how the similar information can lead to error and it also shows how the effort required during encoding affects the memory formation.

1.2 Hypotheses for Report 2

We focus our analysis on four specific hypotheses:

- **H1: Boundary Disruption Effect:** Recognition accuracy is expected to remain stable within events (Pre $\approx$ Mid) but decline sharply immediately following an event boundary (Post), reflecting a boundary-induced disruption in memory performance.
- **H2: Cognitive Load Effect:** Recognition accuracy is expected to differ across task conditions, with higher cognitive load leading to lower overall performance compared to lower load conditions.
- **H3: Doppelgänger Effect:** Highly similar lure items are expected to be more difficult to distinguish from target items, resulting in lower discrimination accuracy compared to less similar lures.
- **H4: Encoding Update Cost:** Reaction time is expected to increase immediately following an event boundary, reflecting the cognitive cost of updating the active event model.

2 Methods

2.1 Experimental Design

Our study included 160 participants (final analysis was only on $n = 159$) and these participants were assigned to one of the three groups based on the encoding tasks performed by them. The item only group ($n = 56$) answered whether each object was indoor/outdoor. The task only group ($n = 53$) answered whether the object will fit in a shoebox and the both item & task group ($n = 50$) performed both of these tasks simultaneously.

During the encoding phase the participants viewed 280 images in 40 blocks of 7 items each, in that they had a scene change or task shift or both at each block’s transition which acted as event boundary. The items were classified by the position like position 1 (post-boundary item), position 4 (mid item, baseline), position 7 (pre-boundary item). During the MST or the test phase the participants were shown 150 images as targets, lures (across five similarity bins, Bin 1 = most similar), and foils as “Old”, “Similar”, or “New” respectively.

2.2 Variables Computed

Bias-corrected memory indices that were computed per participant at each boundary position are as follows:

$$\begin{aligned}\text{REC} &= P(\text{“Old”} \mid \text{Target}) - P(\text{“Old”} \mid \text{Foil}) \\ \text{LDI} &= P(\text{“Similar”} \mid \text{Lure}) - P(\text{“Similar”} \mid \text{Foil})\end{aligned}$$

REC captures the recognition accuracy meaning if the participant is able to discriminate between New and the Old images and LDI captures pattern separation ability meaning if the participant is able to distinguish between Similar (Lures) or New images. We have subtracted $P(\text{Old})$ so as to remove the bias for those participants who responds Old for every image. Along with this Retrieval reaction time (RT) was also recorded for each test trial.

2.3 Statistical Analysis

All analyses were conducted in R using appropriate statistical modeling packages.

- **Linear Mixed-Effects Model (H1):** In LMM we have considered Boundary Position (within-subjects factor with three levels: Pre, Mid, Post) as a fixed effect and participant was random intercept and we used it to test the main effect on REC. A Type III ANOVA was used to test for overall differences, followed by planned comparisons.
- **Mixed ANOVA (H2):** We used 3 x 3 mixed ANOVA to examine interaction between Task group (between-subjects factor) and Boundary position (within-subjects factor). This type of comparison allowed us to test whether there was an effect of the event boundaries on the memory performance across different levels of cognitive load.
- **GLMM - Logistic Regression (H3):** GLMM was used with logistic link function. Lure similarity and task group was used as fixed effects, participant as a random intercept and trial-level accuracy was used as the outcome. This tested whether perceptual similarity and the cognitive load influences discrimination performance.
- **Linear Mixed-Effects Model (H4):** We used LMM with boundary position (Pre, Mid, Post) as fixed effects and participants as a random intercept to analyse the reaction time during the encoding phase. This was followed by planned comparisons to measure the change in reaction time at post-boundary location.

3 Data Integration and Validation Diagnostics

Before conducting the main analyses, we performed some preliminary checks to ensure data quality and consistency across different conditions. In figure 1 we present key descriptive distributions used to validate the dataset.

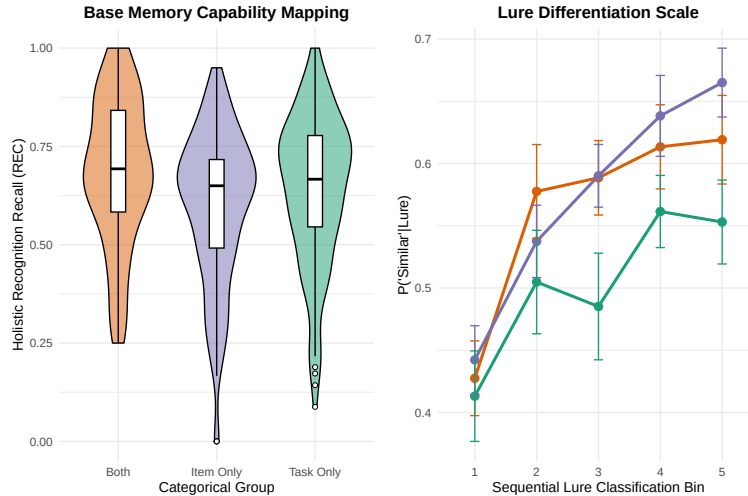


Figure 1: Benchmark distributions illustrating data structure. Left: Violin plots of Recognition Memory (REC) across task groups. Right: Line plots showing the relationship between lure similarity (Lure Bin) and the probability of responding “similar” to lure items.

The left panel’s violin plot shows that the overall REC performance is relatively consistent across three groups. There were no extreme outliers or irregular distributions which indicated that the dataset is suitable for further statistical analysis.

In the right panel we can see the lure discrimination curves which displays a clear increasing trend across all 5 lure bins, here Bin 1 is representing highly similar lures which are very close to the Old items and Bin 5 represents, less similar lures means which are more distinct from the original items. From the figure we can see that as the bin number increases the probability of the participant correctly identifying lure bin as "similar" also increases. This pattern is consistent across all the groups.

From the analysis we can confirm that the dataset exhibits stable and interpretable structure which can be used as a reliable source for our subsequent inferential modelling.

4 Results: Boundary Disruption Effect (H1)

To test whether event boundaries disrupt recognition memory, we used LMM and we have considered Boundary Position (within-subjects factor with three levels: Pre, Mid, Post) as a fixed effect and participant was random intercept and we used it to test the main effect on REC. A Type III ANOVA was used to test for overall differences, followed by planned comparisons.

Hypothesis (H1): Recognition accuracy remains stable within events ($\text{Pre} \approx \text{Mid}$) but drops after a boundary (Post), reflecting a boundary-induced disruption in memory performance.

Table 1: Type III ANOVA: Main Effect of Boundary Position on REC (LMM, H1).

Effect	df (num)	df (den)	F	p-value
Boundary Position	2	318	16.13	< .001

A highly significant main effect of boundary position was observed for REC ($F(2, 318) = 16.13$, $p < .001$). Post-boundary accuracy was significantly lower than mid-event accuracy ($t = -4.93$, $p < .001$) which confirms that there is a boundary-related disruption in recognition performance.

Table 2: Planned contrasts for boundary position effect on REC (H1).

Comparison	t -value	p-value
Mid vs Post	-4.93	< .001
Pre vs Post	-3.79	< .001
Pre vs Mid	0.62	.535

Planned contrasts which were performed confirmed that the post-boundary position was significantly lower than both the mid-event ($t = -4.93$, $p < .001$) and pre-boundary ($t =$

$-3.79, p < .001$) positions. Pre and mid did not reliably differ ($t = 0.62, p = .535$), indicating that the disruption is localised to the moment of contextual transition. Descriptively, mean REC was $M_{\text{pre}} = 0.665$ ($SD = 0.181$), $M_{\text{mid}} = 0.665$ ($SD = 0.187$), and $M_{\text{post}} = 0.594$ ($SD = 0.205$).

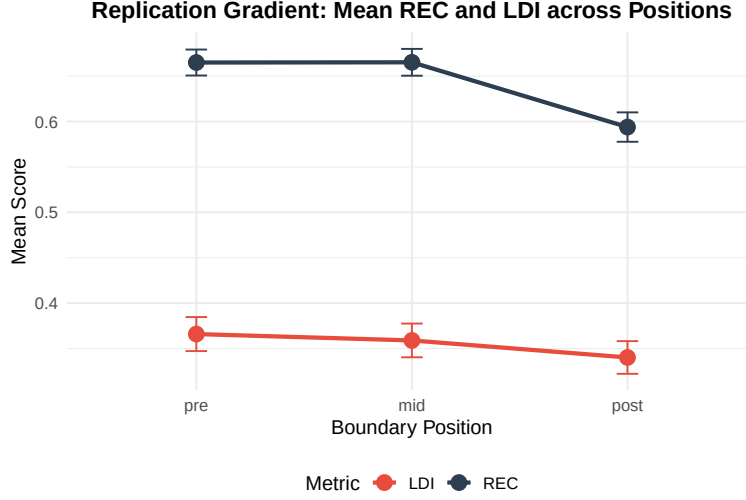


Figure 2: H1 Result: Recognition accuracy by boundary position. Error bars represent ± 1 SE.

Verdict: Supported From the above analysis we can conclude that Recognition accuracy is stable within events but it drops sharply after the boundary and this confirms that there certainly is a disruption in the memory performance which is induced by the boundary.

5 Results: Cognitive Load Effect (H2)

To examine how the cognitive load influences recognition memory we used 3 x 3 mixed ANOVA to examine interaction between Task group (between-subjects factor) and Boundary position (within-subjects factor). This type of comparison allowed us to test whether there was an effect of the event boundaries on the memory performance across different levels of cognitive load.

Hypothesis (H2): Recognition accuracy is expected to differ across task conditions, with higher cognitive load leading to lower overall performance compared to lower load conditions.

Table 3: Results of the Type III Mixed ANOVA examining the effects of Task Group and Boundary Position on recognition memory (REC).

Term	Sum Sq	Mean Sq	Num DF	Den DF	F-value	p-value
boundary_position	0.541	0.270	2	314	16.303	< 0.001
group	0.645	0.323	2	157	4.433	0.0134
boundary_position:group	0.123	0.031	4	314	1.859	0.1174

The ANOVA revealed a significant main effect of task group ($F(2, 157) = 4.433, p = 0.013$), which confirmed that recognition accuracy differs across cognitive load conditions. It also revealed that the participants which were given higher-load task they demonstrated lower overall recognition performance compared to the participants which were in the baseline Item Only condition.

The main effect of boundary position was also significant ($F(2, 314) = 16.303, p < .001$), consistent with H1. But, the interaction between group and boundary position was not statistically significant ($F(4, 314) = 1.859, p = 0.117$), which indicates that even if the cognitive load affects the overall accuracy but it does not amplify the boundary related decline like more decline in Both group than the Item group.

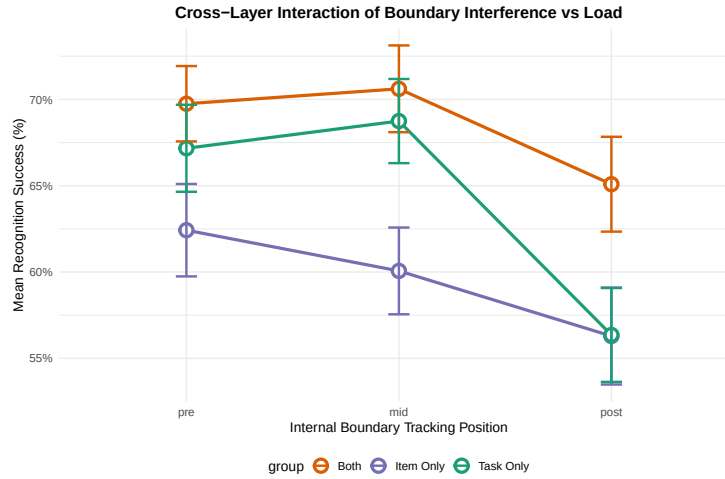


Figure 3: H2 Result: Cognitive Load $\times$ Boundary Position interaction. Higher load reduces overall accuracy, but the boundary disruption pattern is similar across groups.

As shown in Figure 3, the Task Only and Both conditions are exhibiting consistently lower recognition accuracy compared to the Item Only group. This shows that cognitive demands which are due to more complex encoding tasks reduces the memory performance, this is supporting our cognitive load effect.

Verdict: Supported Higher cognitive load significantly reduces overall recognition accuracy.

Since the main effect for the Task group was statistically significant so Bonferroni-corrected post-hoc pairwise comparisons were conducted so as to identify which specific groups differed in the overall recognition accuracy.

Table 4: Bonferroni-corrected post-hoc pairwise comparisons for Task Group main effect on REC (H2).

Comparison	Mean Difference	p (Bonferroni)
Item Only vs Task Only	0.089	0.010
Item Only vs Both Item & Task	0.044	0.441

Task Only vs Both Item & Task	-0.045	0.430
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Post-hoc Result: Item Only participants ($M = 0.685$, $SD = 0.147$) significantly outperformed Task Only participants ($M = 0.596$, $SD = 0.174$; $\Delta M = 0.089$, $d = 0.55$, $p_{\text{Bonf}} = 0.010$), which represents a medium-sized cognitive load effect. The comparisons between Item Only and Both Item & Task ($p_{\text{Bonf}} = 0.441$) and between Task Only and Both Item & Task ($p_{\text{Bonf}} = 0.430$) did not reach any statistical significance after correction. The above values confirms that higher cognitive loads is leading to lower overall recognition accuracy.

6 Results: Doppelgänger Effect (H3)

To examine how perceptual similarity influences memory performance, GLMM was used with logistic link function. Lure similarity and task group was used as fixed effects, participant as a random intercept and trial-level accuracy was used as the outcome. This tested whether perceptual similarity and the cognitive load influences discrimination performance.

Hypothesis (H3): Highly similar lure items (Bin 1) are expected to be more difficult to distinguish from target items than less similar lures (Bin 5), resulting in lower discrimination accuracy.

Table 5: GLMM results showing fixed effects of lure similarity (Lure Bin) on discrimination accuracy.

Parameter Term	Estimate	SE	z-score	p-value
(Intercept) [Item Only]	-0.285	0.133	-2.15	0.031
lure_bin [Item Only]	0.196	0.036	5.41	< 0.001
group (Task Only)	-0.095	0.148	-0.64	0.521
group (Both Item & Task)	-0.155	0.173	-0.89	0.372
lure_bin:task_only	0.042	0.045	0.95	0.343
lure_bin:Both	-0.056	0.052	-1.08	0.282

Result (Supported): The above analysis revealed that there is a strong effect of lure similarity on memory performance ($\beta = 0.196$, $p < 0.001$). Like, participants correctly identified lure item as similar when it was less visually similar to the original item and in contrast to this the participants frequently mistook highly similar bins (Bin 1) as old item, which lead to lower discrimination accuracy.

This pattern was consistent across all task groups. We did not find any significant differences between the groups and there was no reliable interaction between lure similarity and the task condition. This indicated that the effect of similarity on memory is independent of the cognitive load.

This pattern is visualized in Figure 4, which shows a monotonic increase in accuracy across similarity bins.

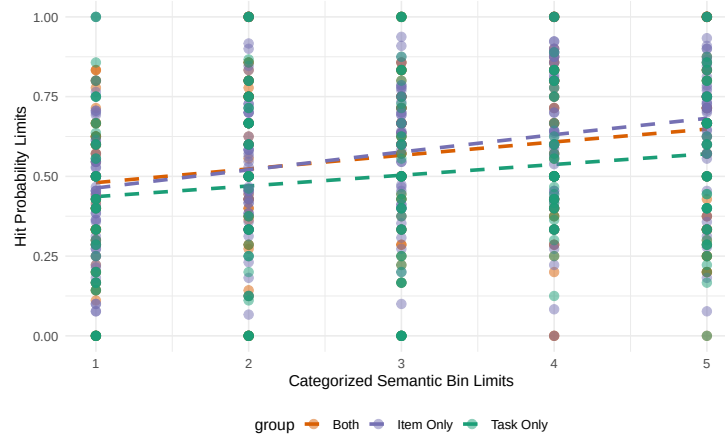


Figure 4: H3 Result: Lure discrimination by similarity bin. Less similar lures (higher bin numbers) are identified more accurately across all groups.

Verdict: Supported From above analysis we can conclude that perceptual similarity is a robust and consistent driver of memory errors, independent of task demands.

7 Results: Encoding Update Cost (H4)

To analyse the cognitive cost of updating the event model during the encoding we used LMM with boundary position (Pre, Mid, Post) as fixed effects and participants as a random intercept to analyse the reaction time during the encoding phase. This was followed by planned comparisons to measure the change in reaction time at post-boundary location.

Hypothesis (H4): Reaction time is expected to increase immediately following an event boundary, reflecting the cognitive cost of updating the active event model.

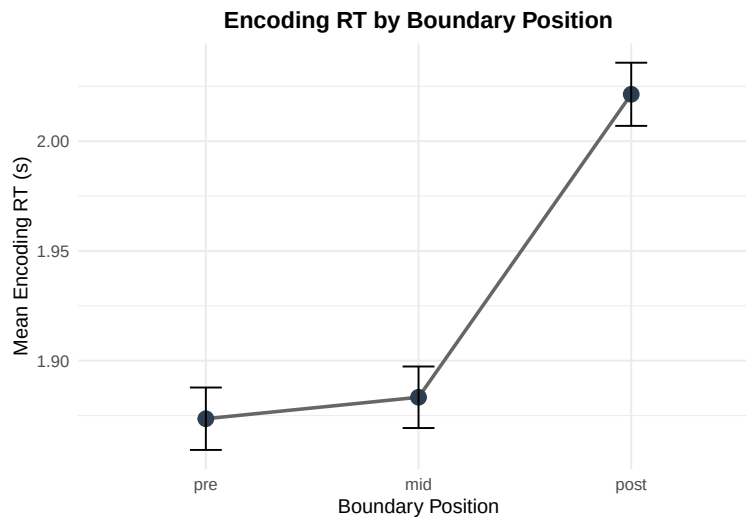


Figure 5: H4 Result: Encoding reaction time by boundary position. RT spikes immediately after boundaries, indicating processing cost of context updating.

Table 6: LMM Results for Encoding Position RT (H4).

Position	Estimate (s)	Std. Error	t-value	p-value
Post (Base)	2.050	0.058	35.51	< .001
Pre vs Post	-0.148	0.019	-7.84	< .001
Mid vs Post	-0.138	0.019	-7.31	< .001

Result (Supported): From above analysis we can see that the encoding reaction time was highest at the post-boundary position. Both pre-boundary and mid-event positions showed significantly lower RT (Pre vs Post: $t = -7.84$, $p < .001$; Mid vs Post: $t = -7.31$, $p < .001$), which confirms that the cognitive cost of updating the active event model is more at the boundary and reduces or vanishes as the event representation stabilises.

8 Discussion and Phenomenological Constraints

The objective of Report 2 was to extend our findings from Report 1 using rigorous statistical models. We applied LMM, Mixed ANOVA, GLMM to understand how the event boundaries and perceptual similarity influences the memory performance in the MST.

8.1 Resolution of Targeted Hypotheses

1. **Boundary Disruption Effect (H1) - Supported:** LMM model which we used revealed that there is a statistically significant main effect of the boundary position on the REC ($F(2, 318) = 16.13$, $p < .001$). The planned contrasts which were performed confirmed that the post-boundary accuracy was significantly lower than the mid-event accuracy ($t = -4.93$, $p < .001$) and pre-boundary accuracy ($t = -3.79$, $p < .001$). Also mid and pre did not differ statistically which shows that this disruption is localised at the transition.
2. **Cognitive Load Effect (H2) - Supported:** The analysis which we performed showed that there was a significant effect of task group ($F(2, 157) = 4.433$, $p = 0.013$) which confirmed that if there is higher cognitive load then it reduces the overall REC, so the participants in the more demanding task conditions performed worse than those in the Item only group.
3. **Doppelgänger Effect (H3) - Supported:** We found a strong and highly significant effect of lure similarity ($\beta = 0.196$, $p < 0.001$) which shows that perceptual change is a primary driver for the memory error. This effect was consistent across all the task groups, which indicates that this effect operated independently of the cognitive load.
4. **Encoding Update Cost (H4) - Supported:** Analysis revealed that encoding reaction time was highest at the post-boundary position, while pre and mid event positions showed significantly lower reaction times ($p < .001$ for both). This confirmed that the cognitive cost of updating an active event model is concentrated at the boundary.

8.2 Theoretical Implications

Above findings aligns closely with the Event Segmentation Theory given by Zacks et al., 2007, in which they proposed that the perceptual systems partitions our continuous experiences into discrete events. Our finding of decline at the post-boundary supports the idea that the memory updating at the event transitions indeed temporarily disrupts the encoding process.

Crucially, the results indicate that this disruption is:

- **Position-specific** (H1): accuracy drops at the boundary itself, not gradually,
- **Load-sensitive at a global level** (H2): cognitive demand lowers overall performance,
- **Complemented by perceptual interference** (H3): driven by stimulus similarity,
- **Manifest during encoding** (H4): RT peaks at the post-boundary position, reflecting active event model updating.

This suggests that boundary-related memory disruption and similarity-based interference operate as **independent mechanisms** shaping memory performance.

8.3 Constraints and Future Directions

There are several limitations which should be noted. First thing is we used discrete boundary positions (Pre, Mid, Post) which simplifies our continuous encoding process. Second is that binary response of the models (correct vs incorrect) may obscure the nuances of the memory dynamics. Finally there was no neural measures which limits our direct interpretation of the underlying cognitive mechanisms.

In the future work we could extent our findings by changing the data collections like incorporating continuous time-modelling, neural data.

8.4 Overall Hypothesis Summary

Table 7: Summary of hypothesis outcomes.

Hypothesis	Description	Status
Boundary Disruption Effect (H1)	Recognition accuracy stable within events but drops sharply at post-boundary	Supported
Cognitive Load Effect (H2)	Higher cognitive load leads to lower overall recognition accuracy	Supported
Doppelgänger Effect (H3)	Highly similar lures are harder to distinguish, lowering discrimination accuracy	Supported
Encoding Update Cost (H4)	RT is highest at the post-boundary position during encoding, reflecting event model updating cost	Supported

9 Conclusion

The above statistical analyses which we performed extends our descriptive findings of Report 1, as this report provides a more precise and statistically grounded understanding of how the event boundaries and the perceptual similarities influences memory performance in MST.

The **Boundary Disruption Effect (H1)** was supported. Analysis confirmed that there was a significant main effect of boundary position on recognition accuracy. Further planned contrasts showed that post-boundary position was significantly lower than the mid-event accuracy. Also pre and mid did not differ statistically. This confirms that memory disruption is localised where there is contextual transition.

The **Cognitive Load Effect (H2)** was supported. Analysis revealed a significant main effect of task group which confirmed that participants which were in higher-load conditions showcased lower overall recognition accuracy. Post-hoc comparisons which were done showed that Item only vs Task only reached statistical significance while the Both Item & Task group showing intermediate performance. The interaction between boundary position and cognitive load was not significant which indicated that while cognitive load reduces memory globally but the boundary disruption pattern is consistent across all groups.

The **Doppelgänger Effect (H3)** revealed that the perceptual similarity is the main driver for memory errors like highly similar lures were frequently mistaken for old items, while less similar lures were correctly identified as lures more often. This effect was consistent across all task groups and was operating independently of cognitive load.

The **Encoding Update Cost (H4)**'s analysis provided evidence from the encoding phase. From the analysis we confirmed that the reaction time was highest at the post-boundary position and it decreased significantly at pre-boundary and mid-event positions this confirmed that the cognitive cost of updating the active event model is concentrated at the boundary.

Finally, our all 4 hypotheses were supported. These findings indicated that the memory performance is shaped by 3 primary mechanisms, first is position specific disruption at event boundaries which is visible in both encoding RT and also during the retrieval accuracy. Second is global reduction in the performance under high cognitive load. This is that there is that there is a strong influence of perceptual similarity on discrimination accuracy. Also more importantly we found out that these mechanisms appear to operate independently, which highlights the distinct roles of event segmentation, cognitive demand and also the stimulus similarity in structuring the memory.

Author Contributions

All members have contributed equally to the project. Srushti Pekamwar performed LMM analyses (H1, H4). Anurag Kaushal calculated key metrics and performed ANOVA (H2). Param Modi did the literature review and performed GLMM analysis (H3). Everyone contributed to data processing, writing, interpretation, final poster and report preparation.

References and Github Reporsitory

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MST_Simpletons GitHub Repository
